

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 11: MUSCL Reconstruction, Slope Limiters, and Second-Order Finite Volume Schemes

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1 Introduction: Where Does Godunov's Scheme Lose Accuracy?

In Lecture 10 we built Godunov's scheme from three steps: **reconstruct** a piecewise constant function from the cell averages, **evolve** it exactly by solving a Riemann problem at every interface, and **average** the result back onto the cells. The scheme is conservative, monotone and TVD, so it never produces spurious wiggles. Its weakness is accuracy. For linear advection we derived its modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\frac{a\Delta x}{2}(1-\nu)}_{\nu_{\text{num}}} \frac{\partial^2 u}{\partial x^2} + \mathcal{O}(\Delta x^2), \quad \nu = \frac{a\Delta t}{\Delta x}. \quad (1)$$

The numerical diffusion $\nu_{\text{num}} = \mathcal{O}(\Delta x)$ smears shocks over several cells and flattens smooth peaks, and on a coarse mesh this damage is severe.

Ask which of the three steps is responsible. The evolution step is exact: the Riemann problem is solved without approximation. The averaging step is exact: it is just the integral conservation law. The only approximation is the **reconstruction step**, where the true profile inside each cell is replaced by a flat constant. A flat constant can represent a smooth function only to $\mathcal{O}(\Delta x)$, and this is exactly where the first-order error comes from.

So the way forward is clear: *reconstruct something better than a constant*. The simplest improvement is a straight line in each cell. This idea is due to Bram van Leer (1979), who named it **MUSCL**: *Monotone Upstream-centred Scheme for Conservation Laws*.

There is one obstacle, and it is an old friend. **Godunov's Barrier Theorem** (Lecture 8) says that a *linear* scheme which never creates new extrema can be at most first-order accurate. So if we choose the slope of the line by a fixed linear formula, we will obtain a second-order scheme that oscillates (in fact we will recover Lax-Wendroff and Beam-Warming). The escape, already hinted at in Lecture 8, is to choose the slope *nonlinearly*, depending on the data. This is the job of a **slope limiter**.

Our roadmap for this lecture:

1. Build a piecewise linear reconstruction that keeps the cell averages exactly.
2. For linear advection, evolve it exactly and show that the three natural slope choices give Lax-Wendroff, Beam-Warming and Fromm, all of which oscillate.
3. Introduce slope limiters (minmod, MC, superbee) that switch off the slope near jumps and extrema.
4. Extend the idea to nonlinear conservation laws: the semi-discrete MUSCL scheme with SSP Runge-Kutta time stepping, and the one-step MUSCL-Hancock scheme.
5. Implement boundary conditions using **ghost cells**, and see why MUSCL needs two layers of them.

2 The Reconstruct-Evolve-Average (REA) Framework

It is useful to write Godunov's scheme in a form where the reconstruction is a separate, replaceable component. Following LeVeque, we call this the **Reconstruct-Evolve-Average (REA)** algorithm. Given cell averages $\{\bar{u}_i^n\}$:

Step 1. Reconstruct a function $\tilde{u}^n(x)$ defined for all x , whose average over every cell is the given $\bar{u}_i^n$.

Step 2. Evolve the conservation law exactly (or approximately) with initial data $\tilde{u}^n(x)$ from t^n to t^{n+1} .

Step 3. Average the evolved function over each cell to obtain $\bar{u}_i^{n+1}$.

Godunov's scheme is REA with a piecewise constant reconstruction. Everything in this lecture keeps Steps 2 and 3 and changes only Step 1. Because Step 3 is the integral conservation law, any scheme built this way is automatically written in the conservative form of Lecture 10:

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \frac{\Delta t}{\Delta x} (F_{i+1/2} - F_{i-1/2}), \quad (2)$$

with $F_{i+1/2}$ the time-averaged flux through the interface $x_{i+1/2}$.

3 Piecewise Linear Reconstruction

3.1 Definition and Conservation of the Cell Average

On a uniform mesh with cells $I_i = [x_{i-1/2}, x_{i+1/2}]$ of width Δx and centres x_i , we reconstruct

$$\tilde{u}^n(x) = \bar{u}_i^n + \sigma_i^n (x - x_i), \quad x \in I_i \quad (3)$$

where σ_i^n is a slope, still to be chosen. Whatever σ_i^n is, the reconstruction has the correct average:

$$\frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} \tilde{u}^n(x) dx = \bar{u}_i^n + \frac{\sigma_i^n}{\Delta x} \underbrace{\int_{x_{i-1/2}}^{x_{i+1/2}} (x - x_i) dx}_{=0} = \bar{u}_i^n. \quad (4)$$

This is essential: the reconstruction adds information about the *shape* of the solution inside the cell, but it never changes the amount of conserved quantity in the cell. Conservation is untouched.

3.2 Accuracy of the Reconstruction

Suppose the data come from a smooth function $u(x)$. Recall from Lecture 10 that $\bar{u}_i = u(x_i) + \mathcal{O}(\Delta x^2)$. If the slope approximates the derivative to first order, $\sigma_i = u_x(x_i) + \mathcal{O}(\Delta x)$, then for $x \in I_i$ (so $|x - x_i| \leq \Delta x/2$):

$$\tilde{u}(x) - u(x) = \underbrace{(\bar{u}_i - u(x_i))}_{\mathcal{O}(\Delta x^2)} + \underbrace{(\sigma_i - u_x(x_i))(x - x_i)}_{\mathcal{O}(\Delta x) \cdot \mathcal{O}(\Delta x)} - \underbrace{\frac{1}{2} u_{xx}(x_i)(x - x_i)^2 + \dots}_{\mathcal{O}(\Delta x^2)} = \mathcal{O}(\Delta x^2). \quad (5)$$

So a piecewise linear reconstruction is second-order accurate, one order better than the piecewise constant one, *provided the slope is a consistent approximation of u_x* .

3.3 Interface States

At each interface the reconstruction now has two values, one from each side:

$$u_{i+1/2}^L = \bar{u}_i + \frac{\Delta x}{2} \sigma_i, \quad u_{i+1/2}^R = \bar{u}_{i+1} - \frac{\Delta x}{2} \sigma_{i+1} \quad (6)$$

For piecewise constant data the jump $u_{i+1/2}^R - u_{i+1/2}^L = \bar{u}_{i+1} - \bar{u}_i$ is $\mathcal{O}(\Delta x)$ in smooth regions. For piecewise linear data the jump is only $\mathcal{O}(\Delta x^2)$, because both one-sided values approximate $u(x_{i+1/2})$ to second order. The Riemann problems at the interfaces therefore become much weaker in smooth regions, and weaker Riemann problems mean less numerical dissipation. At a genuine shock, of course, the jump stays $\mathcal{O}(1)$.

4 Linear Advection: Exact Evolution of a Piecewise Linear Function

For $u_t + au_x = 0$ with $a > 0$ we can carry out the Evolve step exactly: the reconstruction simply translates to the right with speed a . Under the CFL condition $\nu = a\Delta t/\Delta x \leq 1$, the value arriving at the interface $x_{i+1/2}$ at time $t^n + \tau$ ($0 \leq \tau \leq \Delta t$) left the point $x_{i+1/2} - a\tau$ at time t^n , which lies in cell I_i . Hence

$$u(x_{i+1/2}, t^n + \tau) = \tilde{u}^n(x_{i+1/2} - a\tau) = \bar{u}_i^n + \sigma_i^n \left(\frac{\Delta x}{2} - a\tau \right). \quad (7)$$

The exact time-averaged flux is therefore

$$F_{i+1/2} = \frac{1}{\Delta t} \int_0^{\Delta t} a \left[\bar{u}_i^n + \sigma_i^n \left(\frac{\Delta x}{2} - a\tau \right) \right] d\tau = a \bar{u}_i^n + \frac{a}{2} (\Delta x - a\Delta t) \sigma_i^n, \quad (8)$$

that is,

$$F_{i+1/2} = a \bar{u}_i^n + \frac{a}{2} (1 - \nu) \Delta x \sigma_i^n \quad (9)$$

The first term is the Godunov (upwind) flux from Lecture 10. The second term is a **correction** that uses the slope. Substituting into the update (2):

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \nu (\bar{u}_i^n - \bar{u}_{i-1}^n) - \frac{\nu(1-\nu)}{2} (\Delta x \sigma_i^n - \Delta x \sigma_{i-1}^n) \quad (10)$$

4.1 Three Natural Slopes, Three Classical Schemes

There are three obvious ways to approximate u_x in cell I_i from neighbouring averages. Write the backward and forward jumps as

$$\Delta_{i-1/2} \equiv \bar{u}_i - \bar{u}_{i-1}, \quad \Delta_{i+1/2} \equiv \bar{u}_{i+1} - \bar{u}_i. \quad (11)$$

Slope $\Delta x \sigma_i$	Name of slope	Resulting scheme via (10)
0	zero slope	First-order upwind (Godunov)
$\Delta_{i+1/2}$	downwind slope	Lax-Wendroff
$\Delta_{i-1/2}$	upwind slope	Beam-Warming
$\frac{1}{2} (\Delta_{i-1/2} + \Delta_{i+1/2})$	centred slope	Fromm (1968)

Let us verify the Lax-Wendroff row. With $\Delta x \sigma_i = \Delta_{i+1/2}$ and $\Delta x \sigma_{i-1} = \Delta_{i-1/2}$, equation (10) becomes

$$\begin{aligned} \bar{u}_i^{n+1} &= \bar{u}_i - \nu \Delta_{i-1/2} - \frac{\nu(1-\nu)}{2} (\Delta_{i+1/2} - \Delta_{i-1/2}) \\ &= \bar{u}_i - \frac{\nu}{2} (\Delta_{i+1/2} + \Delta_{i-1/2}) + \frac{\nu^2}{2} (\Delta_{i+1/2} - \Delta_{i-1/2}) \\ &= \bar{u}_i - \frac{\nu}{2} (\bar{u}_{i+1} - \bar{u}_{i-1}) + \frac{\nu^2}{2} (\bar{u}_{i+1} - 2\bar{u}_i + \bar{u}_{i-1}), \end{aligned} \quad (12)$$

which is exactly the Lax-Wendroff scheme of Lecture 7. The Beam-Warming and Fromm rows are left as Exercise E1.

Insight: Lax-Wendroff is a Finite Volume Scheme in Disguise

The Lax-Wendroff and Beam-Warming schemes, which we derived in Lectures 7 and 8 from Taylor series, are exactly REA schemes with a piecewise linear reconstruction and a fixed choice of slope. The finite volume viewpoint gives them a geometric meaning: they differ only in *which neighbour is used to measure the slope*.

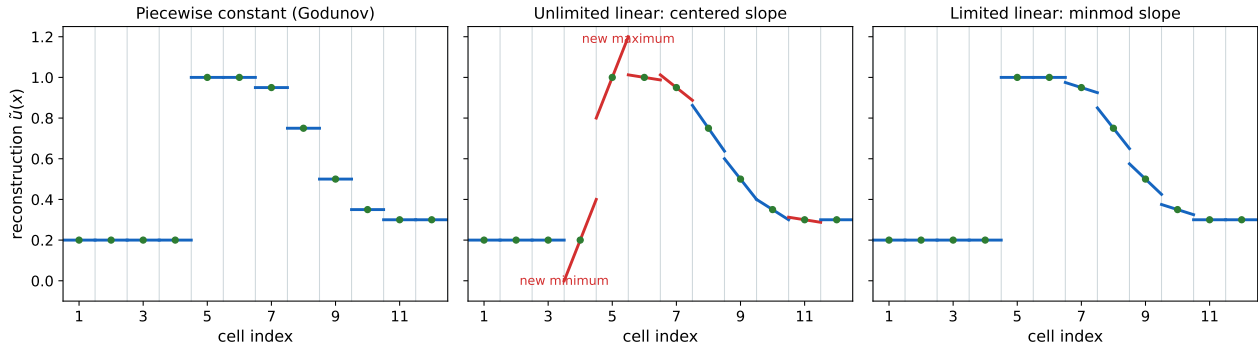


Figure 1: Reconstructions of the same cell averages (green dots). Left: piecewise constant (Godunov). Middle: piecewise linear with the unlimited centred slope; segments in red leave the range of the neighbouring averages and create new extrema next to the jump. Right: piecewise linear with the minmod slope; the slope is switched off next to the jump and at the extremum, and no new extrema appear, while the smooth part is still resolved by sloped segments.

4.2 Why Unlimited Slopes Oscillate

Each of the three slopes is a fixed linear combination of the data, so each scheme is linear, second-order accurate, and by Godunov's Barrier Theorem cannot be monotone. Figure 2(a) confirms this: after one period of a square wave, Lax-Wendroff produces trailing oscillations, Beam-Warming leading oscillations (exactly as predicted by the modified equations of Lecture 8), and Fromm smaller oscillations on both sides.

The reconstruction itself shows where the oscillations are born. Figure 1 (middle panel) uses the centred slope on data with a jump. In the last cell before the jump the centred slope is steep, so the line dips *below* every neighbouring average: the reconstruction has created a new minimum. In the first cell after the jump it rises *above* every neighbouring average: a new maximum. The exact evolution then faithfully transports these artificial extrema, and they appear as wiggles in $\bar{u}^{n+1}$.

5 Slope Limiters

5.1 The Design Principle

Figure 1 suggests a simple rule:

Design Principle of Slope Limiting

Choose σ_i so that the reconstruction inside cell I_i stays between the neighbouring averages:

$$\min(\bar{u}_{i-1}, \bar{u}_i, \bar{u}_{i+1}) \leq \tilde{u}(x) \leq \max(\bar{u}_{i-1}, \bar{u}_i, \bar{u}_{i+1}) \quad \text{for } x \in I_i. \quad (13)$$

In particular, if $\bar{u}_i$ is a local extremum ($\Delta_{i-1/2}$ and $\Delta_{i+1/2}$ have opposite signs), the slope must be $\sigma_i = 0$.

In smooth, monotone regions the limiter should return a slope close to the centred one, so that we keep second-order accuracy. Near jumps and extrema it should reduce the slope, falling back to Godunov's first-order scheme exactly where oscillations would otherwise be born. The resulting slope depends on the data in a nonlinear way, which is precisely what allows us to escape Godunov's Barrier Theorem.

5.2 The Minmod Limiter

Define the **minmod** function, which picks the argument of smallest magnitude if all arguments have the same sign, and returns zero otherwise:

$$\text{minmod}(a_1, a_2, \dots, a_m) = \begin{cases} s \cdot \min(|a_1|, \dots, |a_m|), & \text{if } \text{sign}(a_1) = \dots = \text{sign}(a_m) = s, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

The **minmod slope** is

$$\Delta x \sigma_i = \text{minmod}(\Delta_{i-1/2}, \Delta_{i+1/2}) \quad (15)$$

In words: compare the upwind and downwind slopes, take the gentler one if they agree in sign, and take zero if they disagree.

Proposition: Minmod Creates No New Extrema

With the minmod slope (15), the interface values satisfy

$$u_{i+1/2}^L = \bar{u}_i + \frac{1}{2} \Delta x \sigma_i \text{ lies between } \bar{u}_i \text{ and } \bar{u}_{i+1}, \quad \bar{u}_i - \frac{1}{2} \Delta x \sigma_i \text{ lies between } \bar{u}_{i-1} \text{ and } \bar{u}_i, \\ \text{so the reconstruction obeys (13).}$$

Proof. If $\Delta_{i-1/2}$ and $\Delta_{i+1/2}$ have opposite signs, $\sigma_i = 0$ and there is nothing to prove. Otherwise, say both are positive; then $0 < \Delta x \sigma_i \leq \Delta_{i+1/2}$, so $\bar{u}_i < \bar{u}_i + \frac{1}{2} \Delta x \sigma_i \leq \bar{u}_i + \frac{1}{2} \Delta_{i+1/2} < \bar{u}_{i+1}$. The other bound uses $\Delta x \sigma_i \leq \Delta_{i-1/2}$ in the same way, and the negative case is symmetric. $\square$

Notice that the proof only used $|\Delta x \sigma_i| \leq |\Delta_{i\pm 1/2}|$. In fact any slope with the correct sign and

$$|\Delta x \sigma_i| \leq 2 \min(|\Delta_{i-1/2}|, |\Delta_{i+1/2}|)$$

still keeps the interface values inside the neighbouring range. Minmod is the most cautious choice, and it is also the most diffusive. This leaves room for sharper limiters.

5.3 Sharper Limiters: MC and Superbee

- **Monotonized Central (MC) limiter** (van Leer, 1977): use the centred slope unless it is more than twice either one-sided slope,

$$\Delta x \sigma_i = \text{minmod}\left(2\Delta_{i-1/2}, \frac{1}{2}(\Delta_{i-1/2} + \Delta_{i+1/2}), 2\Delta_{i+1/2}\right). \quad (16)$$

In smooth regions $\Delta_{i-1/2} \approx \Delta_{i+1/2}$, the centred term is the smallest and MC reduces to Fromm's scheme.

- **Superbee limiter** (Roe, 1985): the most compressive choice. With maxmod defined like minmod but picking the *largest* magnitude,

$$\Delta x \sigma_i = \text{maxmod}\left(\text{minmod}(2\Delta_{i-1/2}, \Delta_{i+1/2}), \text{minmod}(\Delta_{i-1/2}, 2\Delta_{i+1/2})\right). \quad (17)$$

With the shorthand $\Delta_- \equiv \Delta_{i-1/2}$ and $\Delta_+ \equiv \Delta_{i+1/2}$:

Limiter	Slope $\Delta x \sigma_i$	Behaviour
Minmod	$\text{minmod}(\Delta_-, \Delta_+)$	Most diffusive; very robust
MC	$\text{minmod}(2\Delta_-, \frac{1}{2}(\Delta_- + \Delta_+), 2\Delta_+)$	Sharp; a good default choice
Superbee	$\text{maxmod}(\text{minmod}(2\Delta_-, \Delta_+), \text{minmod}(\Delta_-, 2\Delta_+))$	Sharpest discontinuities; tends to turn smooth hills into plateaus

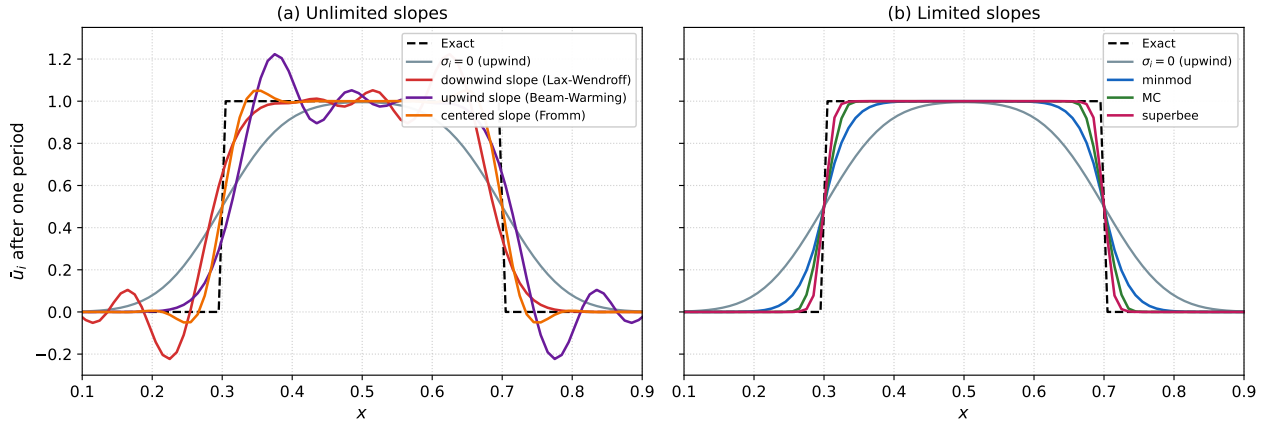


Figure 2: Linear advection of a square wave ($a = 1$, periodic, $N = 100$, $\nu = 0.5$) after one period, using the exact REA update (10). (a) Unlimited slopes: Lax-Wendroff, Beam-Warming and Fromm are second order but oscillate. (b) Limited slopes: minmod, MC and superbee are oscillation-free and much sharper than first-order upwind.

All three return zero at a local extremum. Figure 2(b) shows the result for the square wave: no oscillations at all, and discontinuities far sharper than first-order upwind. Superbee gives the sharpest edges, minmod the most rounded ones, and MC lies in between. Lecture 12 develops the theory behind these choices: why they are TVD, and exactly how far a limiter may go before it starts producing oscillations.

6 Nonlinear Conservation Laws: The MUSCL Scheme

For a nonlinear flux $f(u)$ we can no longer evolve a piecewise linear function exactly: the Riemann problem between two *linear* states has no simple closed-form solution. MUSCL therefore keeps the Riemann solver of Lecture 10 but feeds it the reconstructed interface states (6) instead of the cell averages.

6.1 Semi-Discrete MUSCL (Method of Lines)

Following the Method of Lines of Lecture 9, we first discretize in space only:

$$\frac{d\bar{u}_i}{dt} = \mathcal{L}_i(\bar{u}) \equiv -\frac{1}{\Delta x} \left[\mathcal{F}(u_{i+1/2}^L, u_{i+1/2}^R) - \mathcal{F}(u_{i-1/2}^L, u_{i-1/2}^R) \right] \quad (18)$$

Here $\mathcal{F}(u_L, u_R)$ is any consistent monotone numerical flux from Lecture 10, for example the Godunov flux or the Rusanov flux, and $u_{i+1/2}^L, u_{i+1/2}^R$ come from limited slopes. Setting all slopes to zero recovers the first-order Godunov (or Rusanov) scheme.

6.2 Time Integration: SSP Runge-Kutta

The spatial operator is now second order, so the time integrator must be second order too, otherwise the $\mathcal{O}(\Delta t)$ error of forward Euler would dominate. Just as important, the time integrator must not destroy the non-oscillatory property built into the spatial operator. The standard choice is the **strong stability preserving** second-order Runge-Kutta method (SSP-RK2, also known as Heun's method):

$$\begin{aligned} \bar{u}^{(1)} &= \bar{u}^n + \Delta t \mathcal{L}(\bar{u}^n), \\ \bar{u}^{n+1} &= \frac{1}{2} \bar{u}^n + \frac{1}{2} \left(\bar{u}^{(1)} + \Delta t \mathcal{L}(\bar{u}^{(1)}) \right). \end{aligned} \quad (19)$$

The second stage is an *average* of the old solution and a forward Euler step from $\bar{u}^{(1)}$. So SSP-RK2 is just a convex combination of forward Euler steps, and in Lecture 12 we will see that this

is why it inherits every non-oscillatory property of forward Euler. With a minmod-limited MUSCL reconstruction, a safe time step is

$$\Delta t \leq \frac{1}{2} \frac{\Delta x}{\max |f'(u)|}, \quad (20)$$

that is, a CFL number of 1/2 (proved in Lecture 12).

Algorithm: Semi-Discrete MUSCL Scheme

To evaluate $\mathcal{L}(\bar{u})$:

1. Fill the ghost cells (Section 7).
2. For every cell, compute $\Delta_{i-1/2}$, $\Delta_{i+1/2}$ and a limited slope $\Delta x \sigma_i$.
3. For every interface, form $u_{i+1/2}^L = \bar{u}_i + \frac{1}{2} \Delta x \sigma_i$ and $u_{i+1/2}^R = \bar{u}_{i+1} - \frac{1}{2} \Delta x \sigma_{i+1}$.
4. Evaluate the numerical flux $F_{i+1/2} = \mathcal{F}(u_{i+1/2}^L, u_{i+1/2}^R)$.
5. Return $\mathcal{L}_i = -(F_{i+1/2} - F_{i-1/2})/\Delta x$.

Then advance with SSP-RK2 (19), choosing Δt from the CFL condition at every step.

6.3 The MUSCL-Hancock Scheme

A popular one-step alternative (van Leer, 1984; Toro, Chapters 13–14) evolves the reconstruction for half a time step inside each cell *before* solving the Riemann problems. Denote the two boundary values of the reconstruction in cell I_i by $u_{i,-} = \bar{u}_i - \frac{1}{2} \Delta x \sigma_i$ and $u_{i,+} = \bar{u}_i + \frac{1}{2} \Delta x \sigma_i$.

1. **Predictor.** Advance both boundary values by $\Delta t/2$ using the flux difference across the cell:

$$u_{i,\pm}^{n+1/2} = u_{i,\pm}^n - \frac{\Delta t}{2\Delta x} [f(u_{i,+}) - f(u_{i,-})]. \quad (21)$$

2. **Corrector.** Solve the Riemann problem with these half-step states and update conservatively:

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \frac{\Delta t}{\Delta x} [\mathcal{F}(u_{i,+}^{n+1/2}, u_{i+1,-}^{n+1/2}) - \mathcal{F}(u_{i-1,+}^{n+1/2}, u_{i,-}^{n+1/2})]. \quad (22)$$

The predictor supplies the missing time accuracy (it is a Taylor step $u_t = -f(u)_x$ over $\Delta t/2$), so the scheme is second order in space and time with a single Riemann solve per interface per step. For linear advection, MUSCL-Hancock reproduces exactly the REA update (10) (Exercise E3).

6.4 Results for Burgers' Equation

Figure 3 solves Burgers' equation $u_t + (u^2/2)_x = 0$ with a smooth periodic initial condition that steepens into a shock at $t_b = 1/\pi \approx 0.32$ (compare Lecture 5). At $t = 0.45$ and with only 64 cells, the first-order Godunov scheme clips the peak just before the shock and smears the shock over several cells. Both MUSCL schemes follow the smooth part closely and capture the shock in one or two cells, still without any overshoot. MC is slightly sharper than minmod.

7 Boundary Conditions and Ghost Cells

7.1 How Many Ghost Cells?

On a domain divided into cells $I_1, \dots, I_N$, the update of the first cell needs the flux $F_{1/2}$ at the left boundary. For Godunov's scheme, $F_{1/2} = \mathcal{F}(\bar{u}_0, \bar{u}_1)$ needs one extra value $\bar{u}_0$ outside the domain. For MUSCL, $F_{1/2} = \mathcal{F}(u_{1/2}^L, u_{1/2}^R)$ needs $u_{1/2}^L = \bar{u}_0 + \frac{1}{2} \Delta x \sigma_0$, and the slope σ_0 needs $\bar{u}_{-1}$ as well. So:

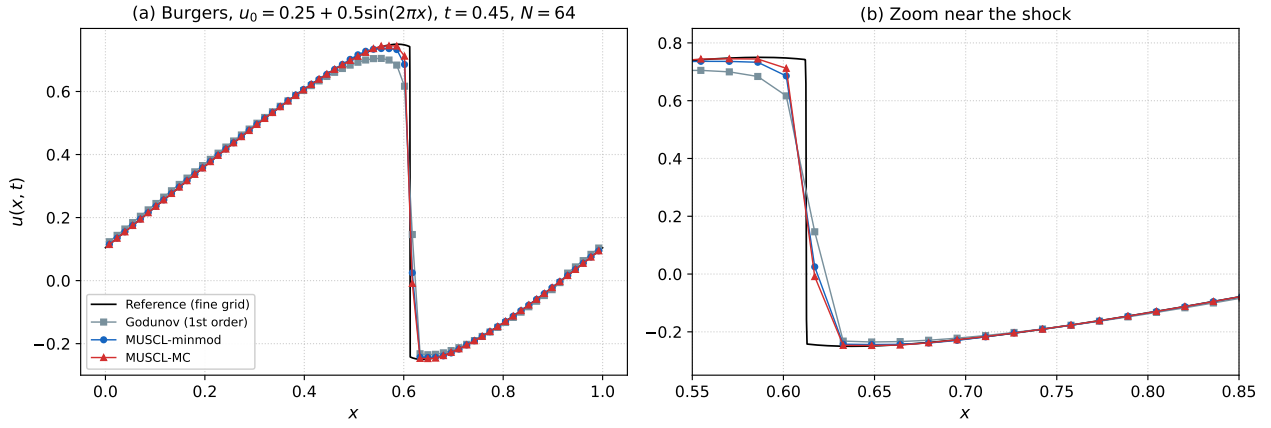


Figure 3: Burgers' equation, $u_0(x) = 0.25 + 0.5 \sin(2\pi x)$ on a periodic domain, $t = 0.45$, $N = 64$, CFL 0.4. First-order Godunov versus semi-discrete MUSCL (Godunov flux, SSP-RK2) with minmod and MC slopes. The reference solution uses MUSCL-MC with $N = 3200$.

Godunov needs 1 ghost cell on each side; MUSCL needs 2 ghost cells on each side.

The general rule: count how far the stencil of the interface flux reaches beyond the interface. **Ghost cells** are fictitious cells outside the physical domain whose values we set at the start of every step (and every Runge-Kutta stage), so that the same interior formula can be used everywhere without special cases.

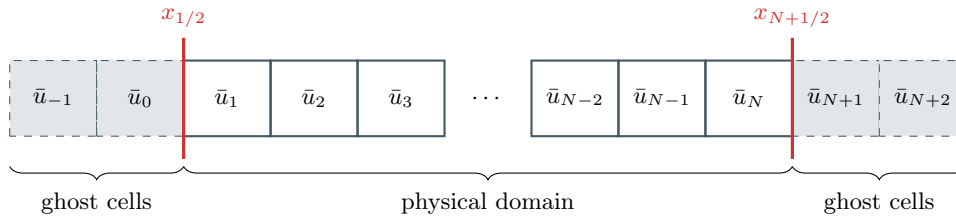


Figure 4: Two layers of ghost cells (shaded) on each side of the physical domain $I_1, \dots, I_N$, as required by the MUSCL stencil.

7.2 Common Boundary Conditions

1. **Periodic.** The domain wraps around, and ghost cells copy the cells at the opposite end:

$$\bar{u}_0 = \bar{u}_N, \quad \bar{u}_{-1} = \bar{u}_{N-1}, \quad \bar{u}_{N+1} = \bar{u}_1, \quad \bar{u}_{N+2} = \bar{u}_2. \quad (23)$$

2. **Transmissive (outflow, zero-gradient).** Waves should leave the domain without reflection. Copy the nearest interior value outward:

$$\bar{u}_0 = \bar{u}_{-1} = \bar{u}_1, \quad \bar{u}_{N+1} = \bar{u}_{N+2} = \bar{u}_N. \quad (24)$$

This makes all interface jumps outside the domain zero, so the boundary Riemann problem is trivial and the outgoing wave simply passes through. The Julia code of Lecture 10 uses exactly this condition.

3. **Inflow (Dirichlet).** If the characteristic at the left boundary points into the domain ($f'(u) > 0$), the boundary value $u(x_L, t) = g(t)$ must be prescribed. Set $\bar{u}_0 = \bar{u}_{-1} = g(t)$ at every stage. At an inflow boundary the solution is determined by the data; at an outflow boundary it must *not* be prescribed, since it is determined by the interior (recall from Lecture 4 that information travels along characteristics).

- 4. Reflective (solid wall).** For systems such as the Euler equations (Modules 5 and 6), a wall is modelled by mirroring: density and pressure are copied ($\rho_0 = \rho_1$, $p_0 = p_1$) while the normal velocity changes sign ($v_0 = -v_1$). The Riemann problem at the wall then produces zero mass flux through it.

Insight: Boundary Conditions Follow the Characteristics

The number of boundary conditions to impose is not a numerical choice: it equals the number of characteristics entering the domain. For a scalar law there is at most one, at the inflow end. Over-specifying (imposing a value at an outflow boundary) produces spurious reflections or boundary layers; ghost cells with zero gradient are the simplest way to let the interior decide.

8 Summary: Godunov versus MUSCL

Feature	Godunov (Lecture 10)	MUSCL (this lecture)
Reconstruction	Piecewise constant: $\tilde{u} = \bar{u}_i$	Piecewise linear: $\tilde{u} = \bar{u}_i + \sigma_i(x - x_i)$, limited slope
Riemann problem input	Cell averages $(\bar{u}_i, \bar{u}_{i+1})$	Interface states $(u_{i+1/2}^L, u_{i+1/2}^R)$
Accuracy	First order everywhere	Second order in smooth monotone regions; first order at extrema and shocks
Time integration	Forward Euler, $\text{CFL} \leq 1$	SSP-RK2 or MUSCL-Hancock
Ghost cells	1 on each side	2 on each side
Linear or nonlinear?	Linear for linear PDEs	Nonlinear even for linear PDEs (escapes Godunov's barrier)

9 Exercises

Exercises for Lecture 11

E1. Classical Schemes from Slopes: For $u_t + au_x = 0$, $a > 0$, use the REA update (10).

- (a) Show that the upwind slope $\Delta x \sigma_i = \Delta_{i-1/2}$ gives the Beam-Warming scheme of Lecture 8.
- (b) Show that the centred slope gives Fromm's scheme, and that Fromm's scheme is the average of Lax-Wendroff and Beam-Warming.
- (c) Repeat the flux derivation leading to (9) for $a < 0$. Which cell does the characteristic come from now, and which slope appears in $F_{i+1/2}$?

E2. Hand Calculation with Limiters: Take cell averages $\bar{u}_1, \dots, \bar{u}_6 = 0, 0, 0.5, 1, 1, 0.4$.

- (a) Compute $\Delta x \sigma_i$ for $i = 2, \dots, 5$ with the centred, minmod, MC and superbee slopes.
- (b) Compute the interface states $u_{i+1/2}^L, u_{i+1/2}^R$ for each choice, and identify which choices create new extrema.
- (c) Explain why all three limiters give $\sigma_5 = 0$.

E3. MUSCL-Hancock for Linear Advection: Let $f(u) = au$ with $a > 0$ and use the upwind flux $\mathcal{F}(u_L, u_R) = au_L$.

- (a) Show that the predictor gives $u_{i,+}^{n+1/2} = \bar{u}_i + \frac{1}{2}(1 - \nu)\Delta x \sigma_i$.
- (b) Deduce that MUSCL-Hancock coincides with the REA update (10).

E4. Ghost Cells:

- (a) A scheme computes $F_{i+1/2}$ from $\bar{u}_{i-2}, \dots, \bar{u}_{i+3}$. How many ghost cells does it need on each side?
- (b) For Burgers' equation on $[0, 1]$ with $u(x, 0) > 0$, which boundary needs an inflow condition and which a transmissive one? What goes wrong if you prescribe u at $x = 1$?

E5. Computational Exercise: Run the code `burgers_muscl.jl` from the course website.

- (a) Compare Godunov, MUSCL-minmod and MUSCL-MC for the moving shock $u_L = 1, u_R = 0$. Count the number of cells inside the shock for each scheme.
- (b) Replace the limiter by the unlimited centred slope. What happens near the shock?
- (c) Replace SSP-RK2 by forward Euler (a single stage). Is the solution still free of oscillations at CFL 0.4?

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